

B.Tech. Degree III Semester Regular/Supplementary Examination January 2023

ME 19-205-0305 METALLURGY AND MATERIAL SCIENCE (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Study the crystal structure, crystal systems, crystallographic planes, crystal imperfections, solidification, and diffusion.
- CO2: Understand the phase rule, phase diagrams, solid solutions, examples of binary solid solutions.
- CO3: Learn about different heat treatment processes for steel and applications of ferrous and nonferrous alloys.
- CO4: Understand different failure mechanisms.
- Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
- PO – Programme Outcome

PART A (Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) What is the importance of Miller indices? Sketch (122) plane and [310] direction in simple cubic structure.	3	3	L1	1	3
(b) Classify crystal imperfections. Differentiate between line dislocation and screw dislocation.	3	3	L3	1	2
(c) Differentiate between carburising and nitriding surface treatment processes.	3	3	L2	1	1
(d) State and explain Gibb's phase rule.	3	3	L2	2	2
(e) What is the difference between failure of a material by creep and by stress rupture?	3	3	L2	4	2
(f) Differentiate between ductile fracture and brittle fracture.	3	3	L2	3	1
(g) What do you mean by super alloys and state their uses?	3	3	L2	3	1
(h) Write short note on Biomaterials.	3	3	L1	4	2

PART B

(4 × 12 = 48)

II. (a) Find the miller indices of a plane that makes an intercept of 1 on the a-axis and 2 on the b-axis and parallel to the c-axis.	6	L5	1	3
(b) Compute the Atomic Packing Factor (APF) for Body Centred Cubic (BCC) and Face Centred Cubic (FCC) crystal structures.	6	L5	1	3
OR				
III. (a) What is interplanar spacing and calculate the interplanar spacing of (220) set of planes of BCC iron if the lattice parameter of iron is 0.2866nm?	6	L5	1	4
(b) Differentiate between homogeneous nucleation and heterogeneous nucleation and cite some typical applications.	6	L2	1	2
IV. Draw the Fe-Fe ₃ C phase diagram and label the phases and briefly explain the various invariant reactions.	12	L2	2	2

OR

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		Marks	BL	CO	PO
V.	(a) Differentiate between hardness and hardenability of steel and explain a test procedure to measure the hardenability of steel.	6	L2	2	2
	(b) Draw a continuous-cooling transformation diagram (CCT diagram) for eutectoid plain-carbon steel. How does it differ from eutectoid isothermal transformation diagram (TTT diagram) for plain-carbon steel?	6	L3	2	3
VI.	(a) Explain the difference between cold working and hot working.	6	L2	3	2
	(b) Explain Griffith's theory of brittle fracture.	6	L3	3	3
OR					
VII.	(a) Explain elastic, anelastic and viscoelastic behaviour of materials and give examples.	6	L2	3	2
	(b) What are strengthening mechanisms and explain any two strengthening mechanisms?	6	L2	4	2
VIII.	(a) Write down the applications of ferrous and non-ferrous alloy steel.	6	L3	4	2
	(b) What do you mean by stainless steel and explain the different types of it?	6	L2	4	2
OR					
IX.	(a) Explain with one example:	6	L2	4	2
	(i) Shape memory alloys				
	(ii) Aluminium alloys.				
	(b) Write short notes on the following:	6	L2	4	2
	(i) Titanium alloys and its use				
	(ii) Optical fibres.				

Bloom's Taxonomy Levels

L1 = 7.5%, L2 = 55%, L3 = 15%, L4 = 2.5%, L5 = 15%, L6 = 5%

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ME 19-205-0306 MACHINE DRAWING
(2019 Scheme)

Time: 4 Hours

Maximum Marks: 60

Follow 1st Angle Projection

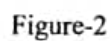
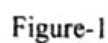
Course Outcomes

On successful completion of the course, the students will be able to:

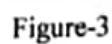
- CO1: Convert a pictorial view in to orthographic view
 CO2: Draw the different views of threaded nuts and bolts
 CO3: Draw the orthographic views of cotter joints, pipe joints, couplings and bearings
 CO4: Draw the assembly drawing of machine parts
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome

		Marks	BL	CO	PO
I.	(a) Pictorial view of a shaft support is shown in Figure-1. Draw the following views: (i) Front view (ii) Top view (iii) End view from right	10	L2	1	1,2
	(b) Draw the three views of a hexagonal headed bolt of size M30. Length of bolt is 100mm and the thread length is 54 mm.	5	L2	2	1,2
OR					
II.	(a) An isometric view of a Knuckle joint is shown in Figure-2. Draw the (i) Bottom half sectional elevation (ii) Top half sectional plan (iii) End view from right side	15	L3	1,3	2
III.	Draw the top half sectional elevation and left side view of flexible coupling shown in Figure-3.	20	L3	1,3	2
OR					
IV.	An isometric view of a Plummer block is shown in Figure-4. Draw to full size: (i) Left half sectional elevation (ii) Top half sectional plan (taking section along A-A) (iii) Full sectional end view (take section along B-B)	20	L3	1,3	2
V.	Disassembled isometric view of a Ramsbottom safety valve is shown in Figure-5. Draw the following views with all parts assembled together (i) Right half sectional elevation Conventional representation of the spring may be followed in the drawing	25	L4	1,4	1,2
OR					
VI.	Draw to the full scale, the right half sectional elevation of the cross head shown in Figure-6 with all parts assembled together.	25	L4	1,4	1,2

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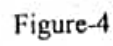


Item	Description	Qty.	Material
1	Fork end	1	M. S.
2	Taper pin	1	M. S.
3	Collar	1	M. S.
4	Pin	1	M. S.
5	Eye end	1	M. S.



ITEM LIST			
Item	Description	Qty.	Material
1	Shaft end	2	Steel
2	Pin (Bolt)	4	M. S.
3	Key	2	M. S.
4	Flange (Bush side)	1	C. I.
5	Bush	4	Rubber
6	Flange (Nut side)	1	C. I.
7	Washer	4	M. S.
8	Nut	4	M. S.

(Continued)



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